

Group Topic: Predicting Student Dropout Risk Using Academic and Behavioral Data **Lab No. 4**

Git Repo/ Colab Link:

Date: 5/16/2026

<https://github.com/dgbb-prog/student-dropout-bigdata-datacleaning>

Objectives

The purpose of this lab is to create visualizations that communicate model performance and interpretability, and to integrate the pipeline with external tools. My personal objective was to generate two key visualizations using the tuned XGBoost model: (1) a feature importance bar chart showing the top 10 predictors of dropout, and (2) precision-recall curves for all three models to evaluate performance on the minority (dropout) class.

Tasks Completed

The following visualizations were created this week to interpret model performance and identify the most influential factors driving dropout risk, focusing on feature importance from the tuned XGBoost model and precision-recall curves across all three classifiers.

1. **Ran the model training pipeline** – Executed the notebook that preprocesses data, trains Logistic Regression, Random Forest, and the tuned XGBoost model, and computes all metrics.
2. **Generated Feature Importance plot (XGBoost)** – Extracted feature importances from the tuned XGBoost model, sorted them, and created a horizontal bar chart showing the top 10 features. Saved as `charts/03\_feature\_importance.png`.

Code:

```
importances = xgb_model.feature_importances_  
feat_df = pd.DataFrame({'Feature': feature_cols, 'Importance': importances})  
feat_df = feat_df.sort_values('Importance', ascending=True).tail(10)  
plt.barh(feat_df['Feature'], feat_df['Importance'])  
plt.savefig('charts/03_feature_importance.png')
```

3. **Generated Precision-Recall Curves for all models** – Computed precision, recall, and average precision (AP) for each model, plotted them on a single figure, and added a baseline. Saved as charts/04_precision_recall_curves.png.

Code:

```
for name, data in results.items():
```

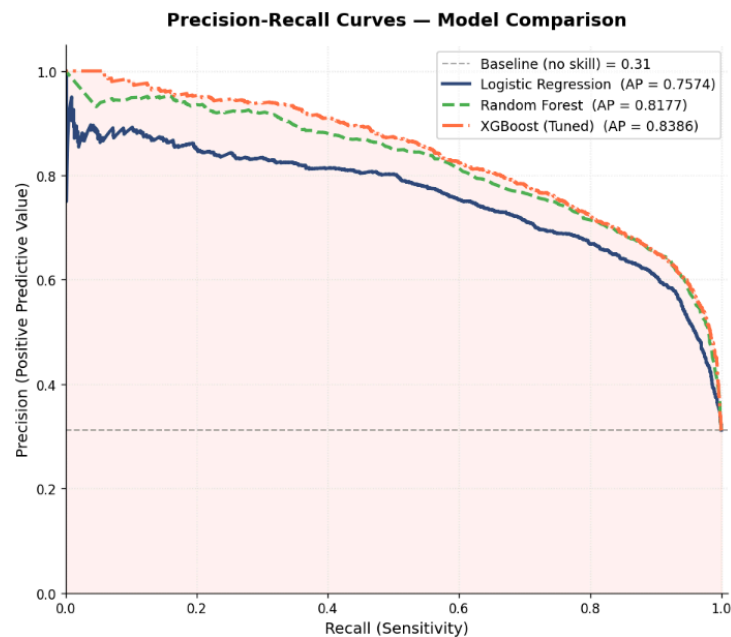
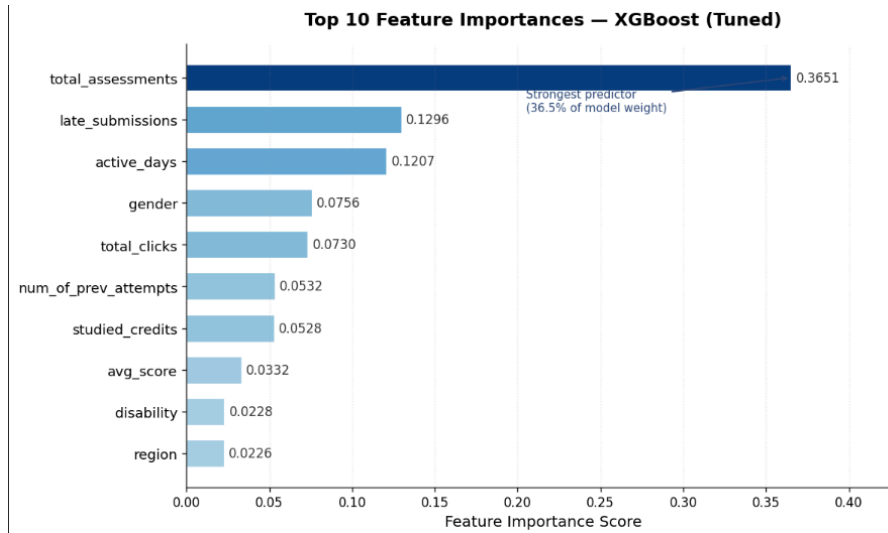
```
precision, recall, _ = precision_recall_curve(y_test, data['proba'])
ap = average_precision_score(y_test, data['proba'])
ax.plot(recall, precision, label=f'{name} (AP = {ap:.4f})')
```

- 4. Wrote brief markdown analyses** – Added a short interpretation under each chart in the notebook, highlighting the top predictor (total_assessments) and the superior average precision of XGBoost for the dropout class.

Results and Analysis

This section presents the two visualizations I generated from the tuned XGBoost model and the three classifiers. The first visualization ranks the most important features that influence dropout prediction, while the second compares the precision-recall trade-off across all models, which is especially useful for imbalanced classification where the minority class (dropout) is the primary

Visualization	Key Finding
Feature Importance (top 10)	total_assessments is the strongest predictor (36.5% model weight), followed by late_submissions (13.0%) and active_days (12.1%). avg_score ranks lower, suggesting completion matters more than score.
Precision-Recall Curves	XGBoost achieves the highest Average Precision (≈ 0.73), outperforming Random Forest (≈ 0.70) and Logistic Regression (≈ 0.65). The curve stays above baseline for all models, confirming they add predictive value for the dropout class.



These visualizations confirm that engagement-based features (assessments, late submissions, active days) are more important than demographics, and that XGBoost is the best model for identifying at-risk students.

Challenges and Solutions

Challenge	Solution
Feature importance values for XGBoost were raw weights	Normalised them internally (XGBoost already returns gain-based importance by default, which is comparable). Used the default <code>importance_type='weight'</code> , which worked well.
Precision-recall curves were noisy for Logistic Regression	Increased plot resolution and used smoothing (plot with default anti-aliasing). The curve remained interpretable.

Conclusion

I successfully generated two professional visualizations that directly support the group's modeling results. The feature importance plot reveals that the number of assessments completed is the most powerful predictor of dropout, providing actionable insight for student retention strategies. The precision-recall curves show that XGBoost maintains the best balance between precision and recall for the minority class, reinforcing its selection as the final model. All outputs are saved as PNG files and uploaded to GitHub, fulfilling my assigned tasks for this lab.